EPIC: Error PredIction and Correction for Power-Efficient Voltage Underscaling Multiply-Accumulate Unit

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***Abstract*—Matrix multiplication dominates the power consumption in compute-intensive applications such as deep neural networks (DNNs), spurring intensive investigations into power-efficient multiply-accumulate (MAC) units. Among the mainstream low-power design methodologies, voltage underscaling can achieve effective power savings yet induce timing errors that may lead to catastrophic accuracy loss. In this paper, we propose an error prediction and correction framework (denoted as EPIC) for arbitrary MAC unit under voltage underscaling, which predicts the timing errors and samples the correct output by using a delay-tunable clock. A prediction bits searching algorithm is proposed to enhance the prediction accuracy with low hardware cost, resulting in up to 100% accuracy. While preserving the accuracy, EPIC achieves up to 52% power savings over the corresponding MAC operating at nominal voltage. With transistor-level optimizations, EPIC incurs only 8% area and 1% power overheads, achieving 100% error correction under a voltage underscaling ratio of 0.74. Compared to state-of-the-art error resilient circuit designs, EPIC consumes 60%-88% less area. Additionally, to achieve the accuracy performance of EPIC in error-resilient applications, we propose a simulation workflow involving precise timing features, enabling an accurate simulation of voltage underscaling MAC in large-scale applications. The experimental results show that, under voltage-underscaling, the MAC with EPIC consumes 11% less power than the one without EPIC, when a same accuracy as exact implementation is required in multi-layer perceptron (MLP).**

***Index Terms*—Power-efficient, timing error resilient circuit, voltage underscaling**

I. Introduction

With a rapid development, DNNs have been proven highly effective across various machine learning applications, including classification, object detection, and generation [1], [2]. To enable efficient hardware deployment, computing hardware like graphics processing unit (GPU) has been tailored to support dominate computations in DNNs. Matrix multiplication is the most basic and key operation for DNNs, especially for the Transformer-based architectures that have emerged as backbone networks in many large language models [3]–[6]. Thus, the implementation of matrix multiplication significantly affects the power efficiency of the entire computing core. While the scale of DNNs continually evolves, the demand on power-efficient matrix multiplication is increasingly high.

In general, matrix multiplication is implemented by using a systolic array consisting of a large number of MAC units [7]. Thus, power-efficient MAC unit has been of particular interest. Numerous design methods have been proposed to reduce the power consumption of MAC from various perspectives, including approximate computing [8], algorithm optimization [9], and voltage underscaling [10]–[12]. Among them, voltage underscaling is promising in reducing power dissipation as it is a quadratic function of the supply voltage. However, aggressive voltage underscaling would incur serious timing errors due to the slow switching of transistor states in low voltage, which may lead to catastrophic accuracy loss in applications [13]. To address this issue, some schemes have been proposed to speed up circuits under low voltage, including redesigning the standard cells [14], leveraging body bias [15] and adopting dual-VDD [16]. Nevertheless, these techniques are tailored to specific circuits, with limited portability and high implementation complexity. A more feasible technique based on error detection and correction (EDaC) has widely been investigated and implemented, which aims to detect the timing errors due to voltage underscaling and recover them.

Many previous works have demonstrated that EDaC and EDaC-like designs can effectively recover timing errors or mitigate the impact of timing errors in DNN accelerators under voltage underscaling [13], [17]–[25]. In [13], the outputs of the processing element (PE) with timing errors are dropped from the sum of product result; this is valid due to the fact that lacking several multiplication results leads to smaller impact than a wrong partial sum. Based on this scheme, [20] adds a compensation circuit to further enhance the accuracy of DNN accelerators under voltage underscaling. [19], [21] propose designs tailored to the characteristics of the inference network. [19] predicts whether errors will occur based on the input activations and raises the voltage if necessary, while [21] identifies weights that are more sensitive to voltage underscaling and maps them to more robust MACs. In [18], a simple razor flip-flop is used to detect and correct the errors due to timing violations. However, using Razor for EDaC, the designs mentioned above fail to detect or recover all the output bits in a MAC due to the area-accuracy trade-off [26]. In [23], rather than Razor, transition detector (TD) is devised to detect timing errors; latches are used for error correction, resulting in a relatively high area overhead.

In this paper, we propose EPIC for arbitrary MAC unit under voltage underscaling, which predicts and corrects the timing errors with negligible area and power overheads. Specifically, the contributions of this work are as follows.

- The EPIC framework is devised for voltage underscaling MAC, where several internal signals (referred to as prediction bits) in an MAC are utilized to predict the timing errors, rather than comparing the outputs of MAC as traditional EDaC does. The errors are then recovered by sampling the correct output by using a delay-tunable clock. Instead of designing the error resilient circuit at gate-level with Razor, we optimize the EPIC-equipped circuit at transistor-level, resulting in 60%-88% less area consumption compared to state-of-the-art designs.
- We propose a searching algorithm for the combination form of prediction bits, aiming to enhance the prediction accuracy with low hardware cost. While maintaining accuracy, EPIC saves up to 52% power for the MAC under aggressive voltage underscaling, compared with the corresponding MAC operating at nominal voltage.
- We propose a timing-accurate simulation workflow to assess the accuracy of voltage underscaling MAC at application level. Different from error injections, the proposed framework involves precise timing features of circuits, achieving the same results as

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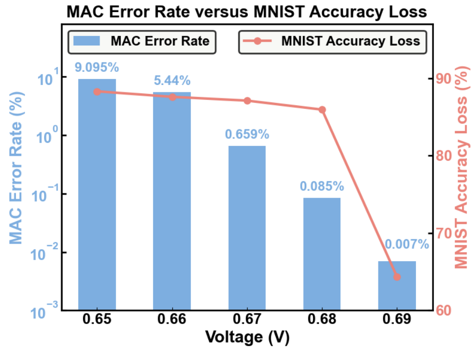


Fig. 1: Error rate in a single low-voltage MAC versus accuracy loss on MNIST classification.

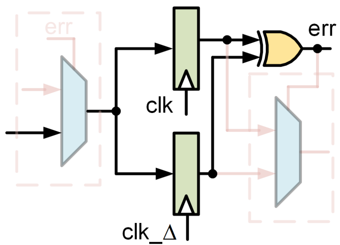


Fig. 2: Schematic of Razor flip-flop.

the corresponding analog/mixed-signal (AMS) simulation while saving time by more than three orders of magnitude. To the best of our knowledge, this is the first work to perform accurate simulation in large-scale applications for voltage underscaling MAC equipped with transistor-level resilient circuits.

II. Background and Motivation

A. The Impacts of Timing Errors in MAC on Applications

To investigate the relationship between the timing errors of MACs and the application accuracy, we utilize a commonly used systolic array to perform MLP inference on the MNIST dataset. Each MAC is synthesized with CMOS 28nm standard cell library using multiplier and adder from Synopsys DesignWare [27], [28]. The error rate of the MAC under each voltage is obtained by providing 300,000 combinations of random numbers as the inputs using VCS simulation. The nominal voltage of the library is 0.9V, corresponding to an MLP inference accuracy of 98.3%.

As shown in Fig. 1, the accuracy loss reaches up to 64.34% even though the error rate in the MAC is only 0.007%. A possible reason can be that, the rarely occurred timing errors lead to significant large error distances, e.g., causing a correct negative result to be sampled as a large positive value, which severely impacts the final outcome. As the voltage continues to decrease, the error rate of MAC rises significantly, while accuracy loss of MLP remains around 87%. At this point, the systolic array can no longer perform reliable computation, which necessities the EDaC for MAC.

B. Razor flip-flop

Razor flip-flop has widely been utilized for EDaC due to its ease of implementation [13], [17]–[21]. As shown in Fig. 2, the key of Razor is borrowing time from the next cycle by double-sampling with two flip-flops, one driven by a normal clock and the other by a delayed clock. The interval between the normal clock and the delayed clock is defined as the detection window. An XOR gate is utilized to compare the outputs of the two flip-flops, generating an error signal as a symbol of timing error. This error signal can then be further used for error processing, e.g., [13] deploys a multiplexor (MUX) before the output register to determine whether to bypass the multiplier, while [18] puts a MUX after the output register to directly obtain the correct result. However, the area overhead caused by the extra flip-flop, MUX and XOR gate will be excessively high if Razor is employed for each bit of MAC outputs, leading to significant area and power consumption for the systolic array. Given this area-accuracy trade-off, a prevalent method is employing Razor for significant bits of MAC outputs instead of all bits [13], [18], [20], which compromises the effectiveness of EDaC.

C. Limitations of Error Injection-Based Simulations

Different from those at nominal voltage, timing specifications of the standard cells under aggressive voltage underscaling are not provided in the library; thus, application-level timing simulation by importing standard cell library for low-voltage circuits is not available. A common methodology for simulating low-voltage circuits at application-level is inserting errors to nominal implementations with specific error rate that is extracted from HSPICE simulation [13], [20], [21]. However, the error due to timing violations highly depends on the current and previous input patterns. Thus, the injected errors as per simple error rate can hardly reflect the timing performance of the circuits under voltage underscaling, as clearly illustrated in Fig. 3.

Fig. 3 diagrams how a timing error happens in a 4 $\times$ 4 array multiplier, where pp1 to pp4 denote the partial product vectors generating by AND gates. The initial inputs are assumed to be 0011 and 1101, resulting in the corresponding accurate output of 00100111. Subsequently, the inputs change to 0101 and 0010; the partial products change accordingly after an AND operation at time t_1. Then, after the time for completing a full addition (FA), at time t_2, sum1-3 and carry1-3 are updated. If the output is sampled between t_1 and t_2, the obtained multiplication result for 0101 and 0010 becomes 00100110. At this time, the error due to timing violation is 28. Similarly, the timing error is 32 when the output is sampled between t_2 and t_3. These indicate that the error distance caused by timing violations is closely related to the internal signal states of a circuit and may persist until the timing margin is satisfied. Computing 0101 $\times$ 0010 requires additional time due to the logic 1 left in the fifth output bit from the previous operation. Thus, timing errors and sampled values depend on both current and previous input patterns as well as prior outputs.

III. EPIC Design

A. Design Overview

Fig. 4 illustrates the MAC unit equipped with the proposed EPIC, where the inputs are 8-bit and the output is 24-bit. In this design, TD is utilized to predict the timing errors by detecting the flips of the logical combination of prediction bits that are selected by a searching algorithm. When a timing error occurs, a delay-tunable clock "clk_Δ" switched by the MUX is used to sample the adder result; otherwise, the nominal clock "clk" is utilized. When correcting the timing errors, the inputs of the adder, including the output from the previous MAC and all bits of the multiplier output (assuming the multiplier completes computation before the rising edge of the clock), are maintained by the transmission gates (TGs) to pad the short paths, preventing the correct data from being overwritten. Unlike EDaC designs using Razor, the timing error correction in this design covers all output bits of the MAC.

B. Prediction Bits Searching for Error Detection

The motivation for error prediction in EPIC lies in the high correlation between the flips of certain internal signals in MAC and timing errors. While sampling the inputs and output at rising edge of the clock, the flips of the more significant output bits in the multiplier and the carry bits in the adder during the negative clock phase are more likely to indicate a long critical path delay in the MAC. However, it is of great difficulty in identifying proper combination of signals for timing error prediction, with low hardware overhead. In this work, we propose

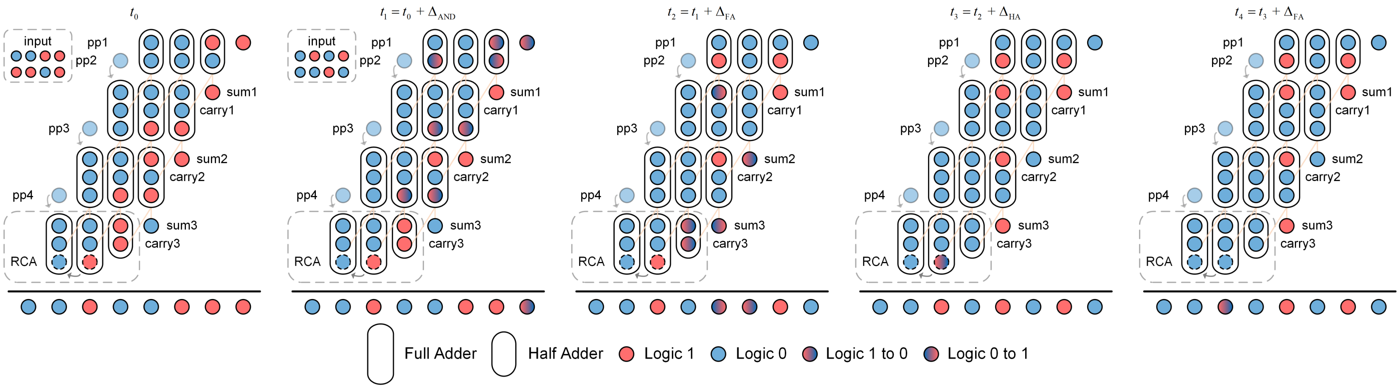


Fig. 3: Diagrams of how a timing error happens in a 4 × 4 array multiplier.

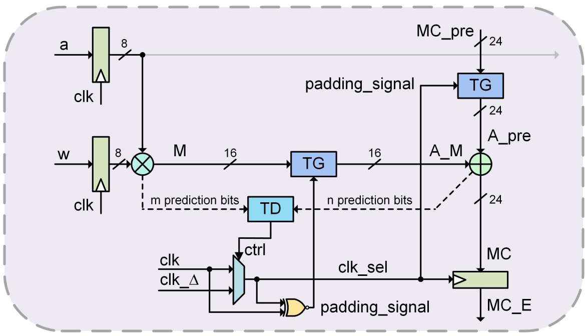


Fig. 4: Schematic of MAC equipped with EPIC.

$Flip\_sta$ (S_1 … S_{34})

case	S_1	S_2	S_3	S_4	⋯	S_{34}	t_err	t_cor
1	0	0	1	1	⋯	0	1	0
2	1	0	1	0	⋯	0	1	0
3	1	0	0	0	⋯	0	0	1
4	1	0	0	1	⋯	1	1	0
5	0	0	0	1	⋯	0	0	1
⋮	⋮	⋮	S_3 and S_{34} can be utilized as prediction bits					⋮

Fig. 5: An example for prediction bits searching.

a searching algorithm to identify the optimal combination form with the fewest internal signals in an arbitrary MAC design, while achieving high prediction accuracy for timing errors.

As shown in Algorithm 1, the inputs include the flip statistics ($Flip\_sta$) of internal signals on the critical path and the corresponding labels for timing errors (t_err), and timing correctness (t_cor), which are obtained based on Synopsys VCS simulations with precise timing features inserted. $Flip\_sta$ records the flipping behaviors of each selected signal in the negative clock phase, based on 300,000 random input patterns. t_err and t_cor label the output state for each input pattern, denoting the occurrence and non-occurrence of timing errors, respectively.

First, the number of signals num to be used is restricted within the range of 1 to max_comb. To shorten the computation time and guarantee a low hardware cost, max_comb is set to 6 in this work. Next, num signals are selected from the candidate signals recorded in $Flip\_sta$; all possible logical combinations of these signals (using only AND and OR gates) are then generated. If a logical combination can predict all timing errors, its prediction accuracy for timing correctness is calculated based on the misprediction rate for t_cor, as shown in line 7. Through iterative loops, the best combination of signals S_{opt},

Algorithm 1: Prediction Bits Search

Input: $Flip\_sta$, t_err, t_cor
Output: S_{opt}, C_{opt}, acc_opt

1 **for** $num \in [1, max\_comb]$ **do**
2   *// Generate all combinations of num bits in Flip_sta*
3   **for** *each selected signal set* S_{curr} **do**
4   *// Generate all logical combinations for* S_{curr}
5   **for** *each logical combination* C_{curr} **do**
6   **if** $t\_err \subseteq pred\_err$ **then**
7                 $acc\_curr = 1 - (|pred\_err| - |t\_err|)/|t\_cor|$
8   **if** $acc\_curr > acc\_opt$ **then**
9                     $S_{opt} = S_{curr}$
10                     $C_{opt} = C_{curr}$
11                     $acc\_opt = acc\_curr$
12   **end**
13   **end**
14   **end**
15   **end**
16 **end**

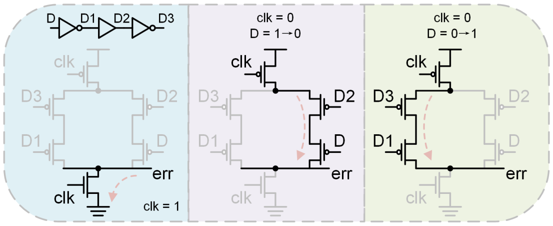
along with the corresponding logical combination C_{opt} and prediction accuracy acc_opt are obtained. Fig. 5 gives a simple example of how prediction bits are selected based on the algorithm. Among the five hypothetical cases, the OR operation of S_3 and S_{34} can predict all timing errors and timing correctness. In contrast, other combinations either fail to predict all timing errors or exhibit a high rate of prediction errors for timing correctness. Fig. 7b illustrates how EPIC predict and correct the timing errors utilizing the combination of prediction bits.

C. Transition Detector Design

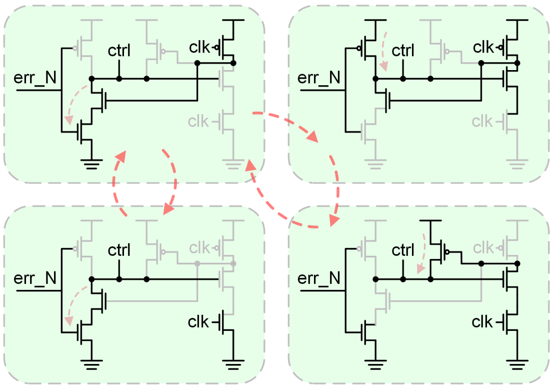
To monitor the flips of the selected prediction bits, TDs are inserted before the adder. Inspired by [23], we propose a power-efficient and low-latency TD consisted of 14 transistors with its detection window positioned in the negative clock phase to enhance the correlation between prediction bits and timing errors.

As shown in the top left of Fig. 6a, the signal D to be monitored is fed to an inverter to generate an inverted signal D1; D1 is then delayed by a buffer, generating D2; a delayed D is finally achieved by inverting D2, denoted as D3. During the positive clock phase, the node “err” is discharged; thus, the signal transition is not monitored. Fig. 6c shows the timing diagram of the related signals, regardless of the input value or whether it changes, the “err” signal remains 0 during the positive clock phase.

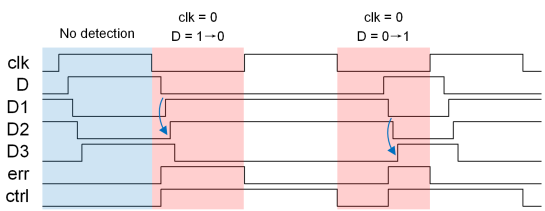
As shown in the middle of Fig. 6a, during the negative clock phase, if D changes from 1 to 0, there is a delay for D2 to change from 0 to 1, causing the transistors on the right to turn on simultaneously, thus creating a charge path from VDD to the node “err”. In this case,



(a) The schematic and working principle of TD.

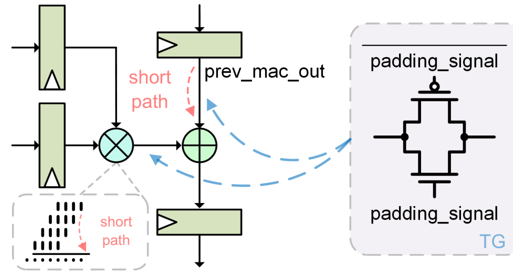


(b) The Schematic and working principle of the control signal generation circuit.

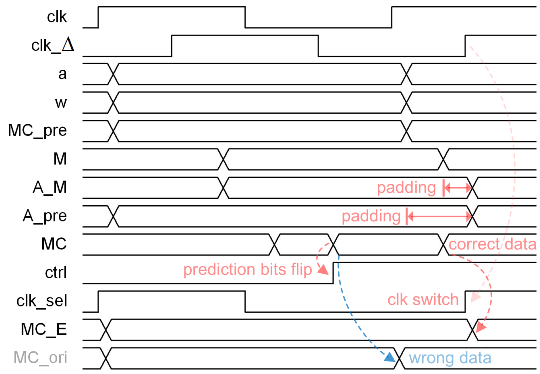


(c) Timing diagram of TD and the control signal generation circuit.

Fig. 6: Schematics, working principle, and timing diagram of TD and the control signal generation circuit.



(a) Illustration of TG for short-path padding.



(b) Timing diagram of TG and EPIC.

Fig. 7: Illustration and timing diagram of TG.

TD generates a positive pulse to indicate an input transition in the detection window, as shown in Fig. 6c. Similarly, the right of Fig. 6a shows the detection of D changing from 0 to 1 during the negative clock phase, another charge path is activated on the left due to the delay between D3 and D1.

When a timing error is predicted to occur, a delayed clock is selected to sample the correct multiplication result. Thus, the prediction result must be available until the next falling edge of the clock to ensure a proper clock switching. However, as the "err" is reset to 0 at the rising edge of the clock, the logical combination of the "err" signals for the prediction bits will disappear at the rising edge of the clock. Thus, a control signal generation circuit is devised to extend the "err" signal until the next clock's falling edge. As shown in Fig. 6b and Fig. 6c, at the beginning of the negative clock phase, the "err" is 0, resulting in the activation of the discharge path from node "ctrl" to the ground. The "err_N" signal, which is the inverted "err" signal, will change to 0 when the prediction bit flips, resulting in the charging of node "ctrl". During the positive clock phase, the feedback path is activated, maintaining "ctrl" at 1 until the falling edge of the clock (bottom right of Fig. 6b). Another possible scenario is that, during the negative clock phase, if the prediction bits do not flip, the "ctrl" signal remains 0 during the positive clock phase, which is shown in the bottom left of Fig. 6b.

D. Short-path padding

The short-path problem is severe in EDaC designs, which is resulted from the activation of short path within the detection window. As shown in Fig. 7a, when the output registers are ready to sample the results at the rising edge of the delayed clock due to a predicted timing error, an early arrival of the adder inputs can happen before the correct data is sampled because the clock of the input registers remains unchanged. The possible early arrived inputs include the output from the previous MAC and the lower bits of the multiplier output.

The short-path problem is intrinsically a hold-time violation, which can be solved by imposing a hold constraint that matches the width of the error detection window on all monitored paths [22]. As EDaC designs based on Razor are commonly devised at gate-level, the traditional method for hold-time violation correction involves inserting buffers. However, this introduces a trade-off between area and accuracy, as larger detection window requires more buffers. Additionally, since the number of padding buffers is hard to adjust in post-fabrication, the detection window can hardly be modified in these designs.

Benefiting from the transistor-level design of EPIC, we adopt TGs for short-path padding [29], with extremely low area overhead and high compatibility with the delay-tunable clock. As shown in Fig. 7a, TG is controlled by a padding signal, which is inserted between the adder and its inputs to prevent the early arrival of the inputs. For the input from the previous MAC, the padding signal is the "clk_sel" generated by the clock switching MUX. This ensures that the output from the previous MAC will be blocked until the rising edge of the delayed clock, while it passes through normally when the normal clock is used. For the input from the multiplier, we insert TG for each bit as we assume that all the multiplications can complete before the rising edge of the clock. In this case, the padding signal is generated by performing an XNOR operation on "clk_sel" and "clk", ensuring a normal propagation of the multiplier output during the negative clock phase. As shown in Fig. 7b, the inputs of the adder are extended to the rising edge of the delayed clock when an error is predicted to happen. As the short-path padding signal is derived from "clk_sel" and "clk", the detection window in EPIC can be easily adjusted by tuning the delayed clock.

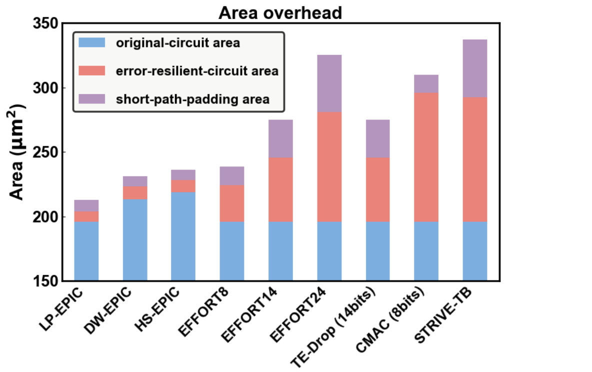


Fig. 8: Area overhead comparison of EPIC-based MAC variants, EFFORT [18], TE-Drop [13], CMAC [20], STRIVE-TB [17].

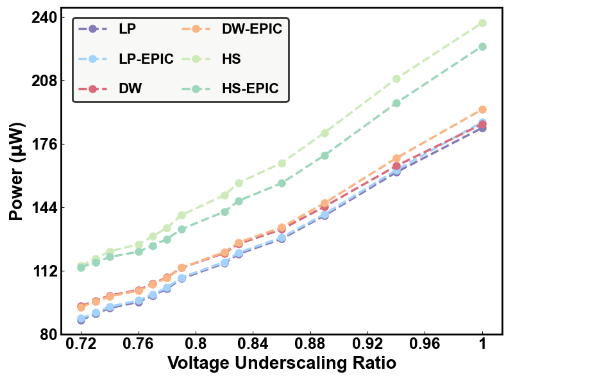


Fig. 9: Power of MACs under different voltage underscaling ratio.

E. Variants of EPIC-Based MAC

To demonstrate the generality of EPIC, we design three MAC structures along with their EPIC-equipped counterparts.

- Low power design (LP): The MAC is completely flattened by EDA tool to minimize power and area, with the multiplication and addition operations merged.
- Design ware design (DW): The multiplier and adder are from Synopsys DesignWare [27], [28], which retains the original multiplier output.
- High speed design (HS): The ripple carry adder (RCA) for the final addition in the multiplier is replaced with a parallel prefix adder based on BK-tree [30], which shortens critical path of the multiplier.

IV. Experiments

In this section, we evaluate EPIC in a standalone MAC unit and in large-scale applications. In Section IV-A, we compare the area, power consumption, and prediction accuracy of MAC units with and without the prediction bits searching algorithm. Additionally, we compare the area of MAC units using EPIC with those employing EDaC from other works. Finally, we evaluate the error correction performance and power overhead of EPIC under different corners. In Section IV-B, we expound upon the detailed work flow for simulation on large-scale application and the corresponding results.

A. Evaluation of MAC Unit

To demonstrate the efficiency of our proposed prediction bits searching algorithm, we evaluate the area, power, and prediction accuracy of the EPIC-based MAC variants, both with and without using the algorithm. For DW and HS, the versions without using algorithm utilize the 6 most significant bits (MSBs) of the multiplier output as the prediction bits while 6 MSBs of the adder carry are used

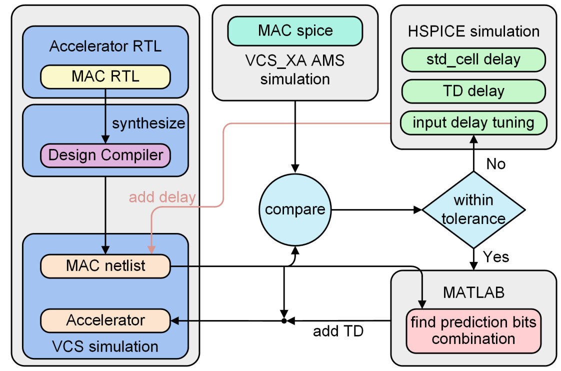


Fig. 10: Timing-accurate simulation workflow for applications.

in LP. Befitting from the searching algorithm, LP utilizes the AND result of two internal signals, DW uses the OR result of the MSB of the multiplier output and two significant bits of adder carries, while HS employs the AND result of the 11^{th} and 12^{th} output bits of the multiplier, followed by an OR operation with the 11^{th} carry bit of the adder. With a 1GHz clock, both LP and DW are tested at 0.67V while HS is tested at 0.66V, with corresponding voltage underscaling ratios of 0.73 and 0.74, respectively.

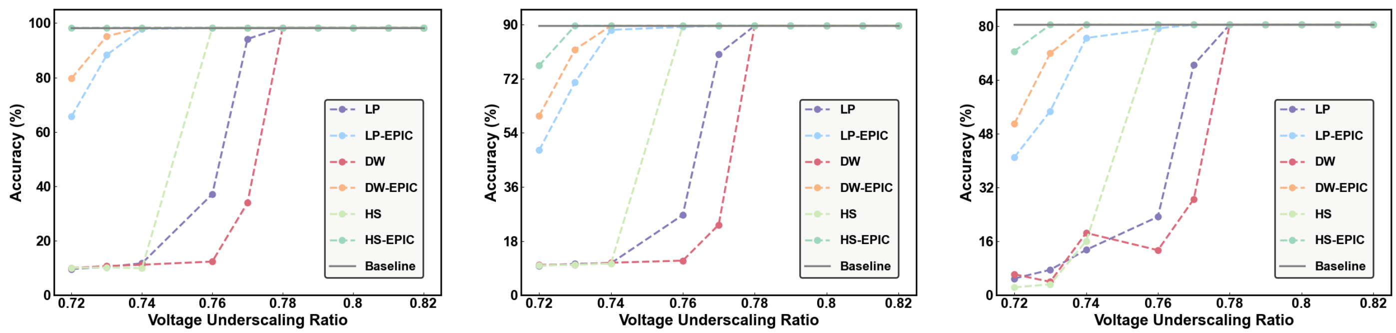
Table I shows the comparison results, where te and tc represent the probabilities of accurately predicting the timing errors and error-free timing, respectively, p denotes the power consumption. It can be seen that EPIC-based MAC variants utilizing the prediction bits searching algorithm achieve 100% prediction accuracy for timing errors, with relatively low power dissipation. This indicates that, benefiting from the algorithm, the EPIC can detect all timing errors, and some predicted timing errors do not happen, resulting in $tc < 100\%$. In contrast, EPIC-based MACs without using the algorithm fail to guarantee 100% prediction accuracy for timing errors, while exhibiting higher power consumption.

To investigate the area overhead of EPIC, we synthesize EPIC-based MAC variants and calculate the area based on CMOS 28nm standard cell library. For designs utilizing Razor, we insert buffers for short-path padding using PrimeTime and report the area overhead. Since these Razor-based designs achieve EDaC by comparing only the outputs of the MAC without requiring the multiplier results, they are implemented using LP to minimize area. Note that the synthesis clock is 1GHz for all the considered designs. As shown in Fig. 8, the area overhead caused by short-path padding increases with the number of protected bits, accounting for up to 14% of the total area. Similarly, the area overhead introduced by Razor-based EDaC circuits is also significant, reaching up to 32% of the total area. In contrast, for circuits utilizing EPIC, the timing-resilient circuits account for up to 4.2% of the total area, while the circuits for short-path padding account for up to 4.3%. Compared to the resilient circuit designs in other works, EPIC consumes 60%-88% less area, with only 8% area overhead in total.

To demonstrate the robustness of EPIC, we also measure the performance of EPIC-based MAC variants at different CMOS technology corners, namely ff, tt, and ss. As shown in Table II, V_{EPIC} and V_{ori} represent the minimum voltages achievable by the circuit with EPIC and the original circuit, respectively, under the condition of no timing errors. P_{overhead} denotes the power overhead of the circuit with EPIC

TABLE I: Comparisons of EPIC-based MAC variants with and without using prediction bits searching algorithm

EPIC variants	with algorithm te (%)	tc (%)	p (μW)	without algorithm te (%)	tc (%)	p (μW)
LP	100	99.5	93.99	98	81.8	99.43
DW	100	87.3	99.2	98.7	92.1	103.9
HS	100	66.1	116.4	93.8	50.3	121.5



(a) MNIST. (b) FMNIST. (c) Reuters.

Fig. 11: Inference accuracy of systolic array on three MLP benchmarks with voltage scaling.

TABLE II: Comparisons of EPIC-based MAC variants at different CMOS technology corners (clk=1GHz).

EPIC Variants	corner	V_{EPIC} (V)	V_{ori} (V)	P_{overhead} (%)
LP	ff	0.57	0.6	1.52
	tt	0.67	0.7	0.70
	ss	0.76	0.8	0.45
DW	ff	0.56	0.6	0.90
	tt	0.67	0.7	-0.44
	ss	0.76	0.81	-1.16
HS	ff	0.56	0.59	-0.50
	tt	0.66	0.69	-1.52
	ss	0.76	0.80	-2.94

at V_{EPIC}. At different corners, EPIC maintains stable error correction performance across all EPIC-based MACs, with a maximum power overhead of only 1.52%. Note that, in some cases, the power overhead of EPIC is negative because the use of short-path padding may reduce adder glitches, resulting in less power than the original circuits. We measure the power consumption of all considered MACs versus voltage underscaling ratio using AMS simulation, as shown in Fig. 9. While preserving the accuracy, EPIC achieves up to 52% power savings over the corresponding MAC operating at nominal voltage.

B. Application

To demonstrate the potential of EPIC in large-scale applications, we deploy three MLP models on the weight-stationary systolic array consisting of 16x16 EPIC-based MACs mentioned in Section III-A. To enable an accurate simulation for voltage underscaling MAC equipped with transistor-level resilient circuits in large-scale applications, we propose a workflow for timing-accurate simulation, as illustrated in Fig. 10. As we focus only on timing errors within MAC, the key to enable an accurate and fast simulation for large-scale applications is building an efficient timing-accurate model for MAC. To this end, the Synopsys Design Compiler is first used to synthesize the RTL implementation of a MAC for constructing accelerator, resulting in the netlist of a MAC. Next, HSPICE is utilized to extract the timing features of standard cells and TDs under voltage underscaling; these timing features are then inserted into the synthesized MAC netlist using Verilog specify statements. Synopsys VCS is then used to simulate the MAC, and the simulation results are compared with those from Synopsys VCS_XA AMS simulation based on real timing features. If significant discrepancies are observed, the input delay of the test stimuli is fine-tuned in HSPICE, as variations in the input delay can lead to differences in the extracted timing features. The comparison is then repeated until the results are nearly identical. Subsequently, the flips of the selected internal signals in MAC and their corresponding timing error occurrence information are recorded and processed in MATLAB, where the prediction bits searching algorithm identifies the optimal prediction bits and their logic combination for TD. Using Verilog primitives, TDs and TGs are modeled and inserted into the MAC netlist. Finally, three pre-trained and quantized MLP models shown in Table III are deployed on the systolic array, and VCS is employed to simulate the inference of the MLP at a clock frequency of 1 GHz. Note that the considered MLP models are quantized into 8-bit integer format. By avoiding numerical simulations with transistor-level SPICE models, the proposed workflow achieves the same accuracy as AMS simulations while saving over three orders of magnitude in simulation time.

TABLE III: MLP models and their error-free accuracy.

Benchmarks		Accuracy (%)
Dataset	Architectures	
MNIST [31]	FC: 784×256×256×10	98.30
FMNIST [32]	FC: 784×256×256×10	89.56
Reuters [33]	FC: 2048×256×256×46	80.45

As shown in Fig. 11, EPIC demonstrates stable error correction performance across all three MAC variants on the three MLP benchmarks. According to Fig. 9, compared to MAC units operating at the minimum voltage required to maintain accuracy, EPIC achieves up to an 11% reduction in power consumption while preserving accuracy. Considering that MACs operating at low voltages encounter IR drop issues, a 10% voltage margin is needed to ensure MACs can maintain accuracy on three MLP benchmarks even with a 10% voltage drop. In this scenario, the operating voltages for the three MAC variants, LP, DW, and HS, should be 0.77V, 0.77V, and 0.75V, respectively. According to Fig. 9, equipped with EPIC, the power consumption of three MAC variants can be reduced by up to 15% while maintaining a 10% voltage margin.

V. Conclusion

In this paper, we propose EPIC, an efficient error prediction and correction framework for MAC units under voltage underscaling. EPIC uses several internal signals of the MAC to predict the timing errors and correct the errors by using a delay-tunable clock to borrow the time margin of computation during the next clock. Moreover, a searching algorithm is proposed to find the optimal prediction bits and their logic combinations for the timing error prediction; it achieves 100% prediction accuracy on timing errors with minimal hardware overhead. Overall, EPIC delivers up to 52% power savings compared to MACs without EPIC at nominal voltage, with only 8% area overheads while ensuring 100% error correction under a voltage underscaling ratio of 0.74. Compared to state-of-the-art designs, EPIC consumes 60%-88% less area. Furthermore, a timing-accurate simulation workflow incorporating precise timing features of circuits is proposed, enabling efficient evaluation in large-scale applications. Experimental results demonstrate an 11% power reduction in MLP inference with EPIC-based MACs under equivalent accuracy constraints.

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